Smart water management including quality, automation and fertigation.

F.L.O.R.A

Fog-assisted Light-integrated Optimized Root Apparatus

Ultra-Low-Cost, closed loop fogponics
for zero-waste agriculture



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PROBLEM STATEMENT

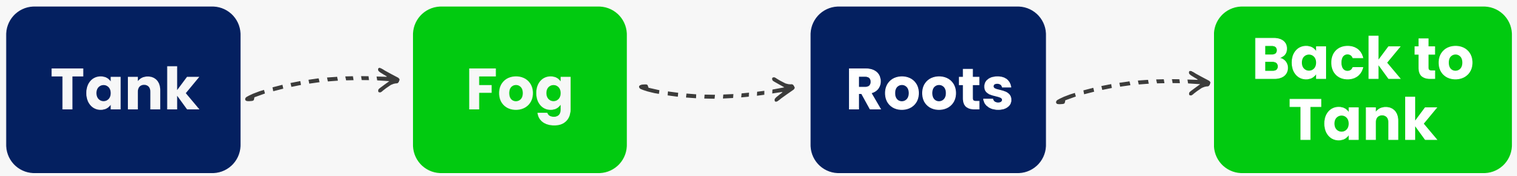
- Agriculture faces serious water scarcity issues.
- Traditional irrigation wastes large amounts of water.
- Aeroponics is efficient but too expensive and complex.
- Low-cost fog systems cause root rot due to heavy droplets.



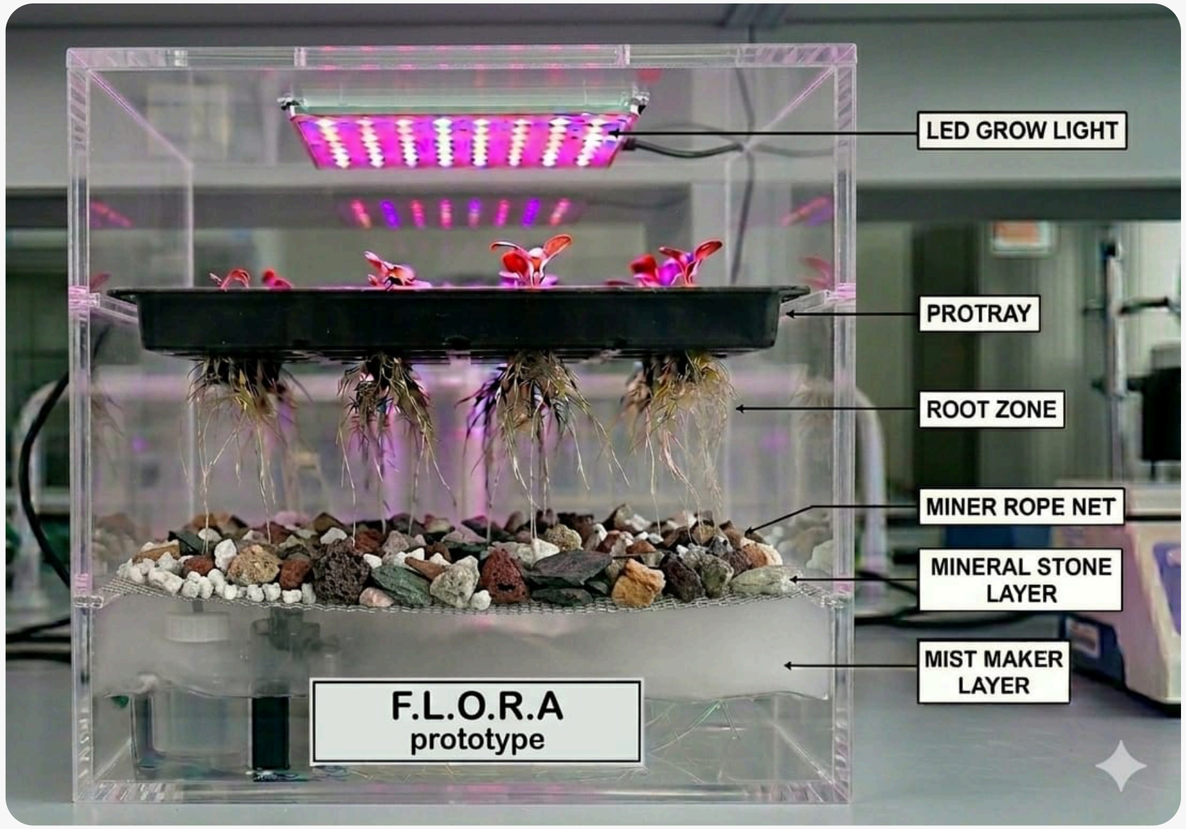


OUR SOLUTION

- We propose F.L.O.R.A., a decentralized and ultra-low-cost fogponics system.
- It uses an ultrasonic mist maker instead of expensive high-pressure pumps.
- A mineral pebble baffle layer ensures only fine “dry fog” reaches plant roots.
- The system operates in a closed loop, minimizing water and nutrient loss.
- It can reduce water usage by up to 95% compared to traditional farming.

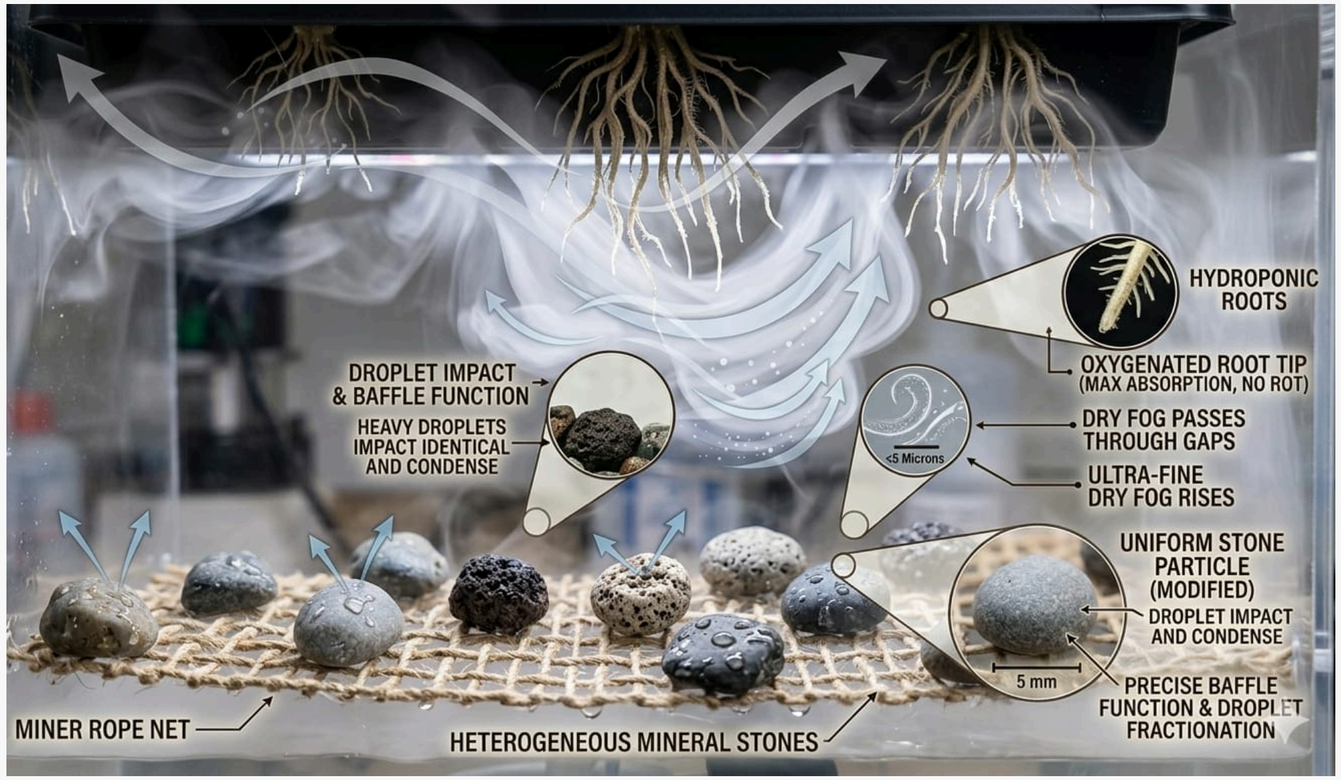


WORKING MECHANISM



- The system operates inside a sealed transparent chamber.
- An ultrasonic mist maker converts nutrient-rich water into a dense fog.
- This fog passes through a layer of mineral pebbles, which filter out heavy droplets.
- Only ultra-fine dry fog reaches the plant roots, ensuring high oxygen availability.
- Unused moisture condenses and returns to the reservoir, creating a continuous recycling loop.

CORE INNOVATION



- The aluminum mesh and mineral pebble layer is our primary innovation. It acts as a passive mechanical sieve.
- It guarantees that roots only receive highly oxygenated "dry fog" (under 5 microns), entirely eliminating the root rot associated with standard fogponic systems.
- Any unabsorbed mist condenses on the enclosure walls or the baffle layer and drips straight back into the reservoir. **Zero water is lost to deep percolation or evaporation.**

DELIVERABLES (PROTOTYPE)

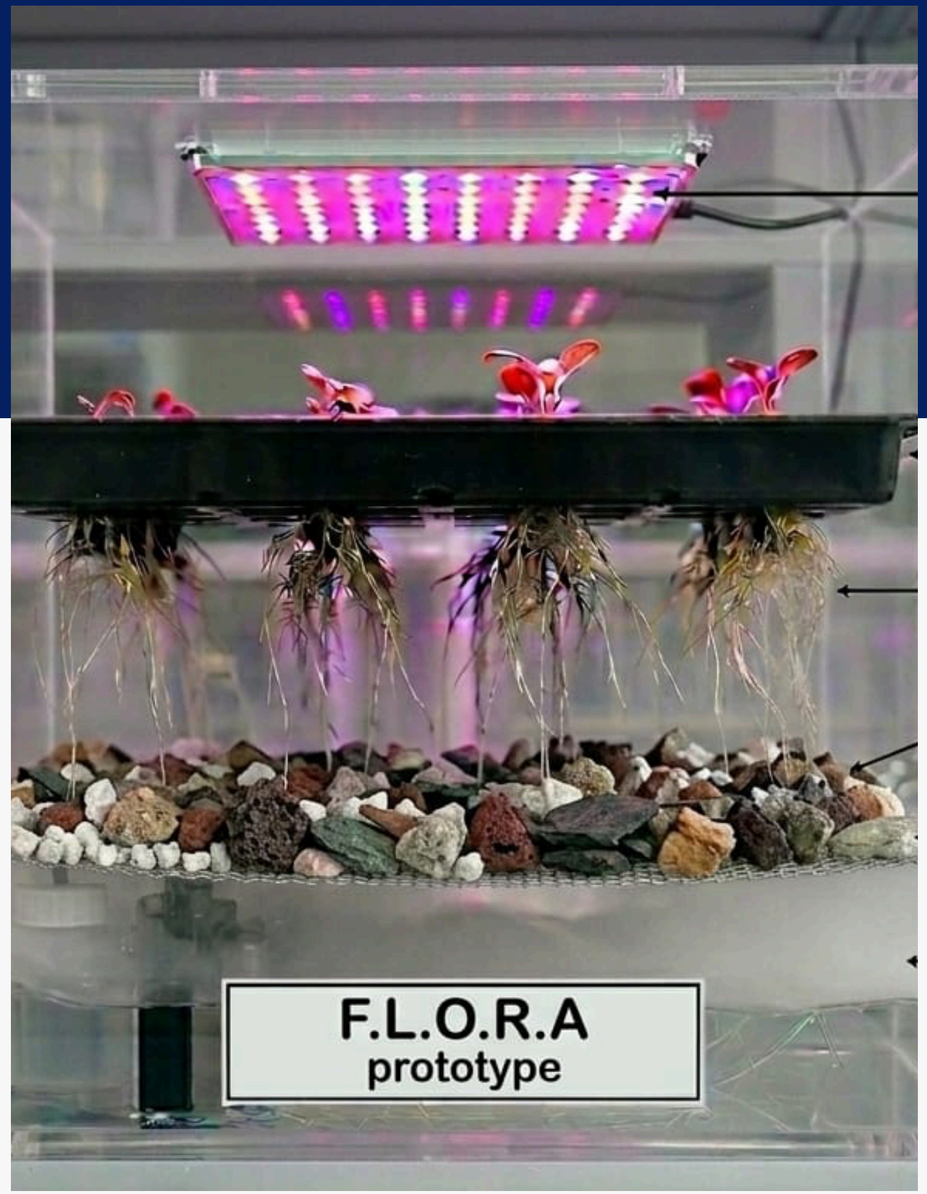
For the live prototype testing and final presentation, our team will deliver:

1. A complete prototype of the F.L.O.R.A. bio-reactor demonstrating the closed-loop fertigation cycle.

2. A live visual demonstration of the mist fractionation process using the mineral pebble baffle layer.

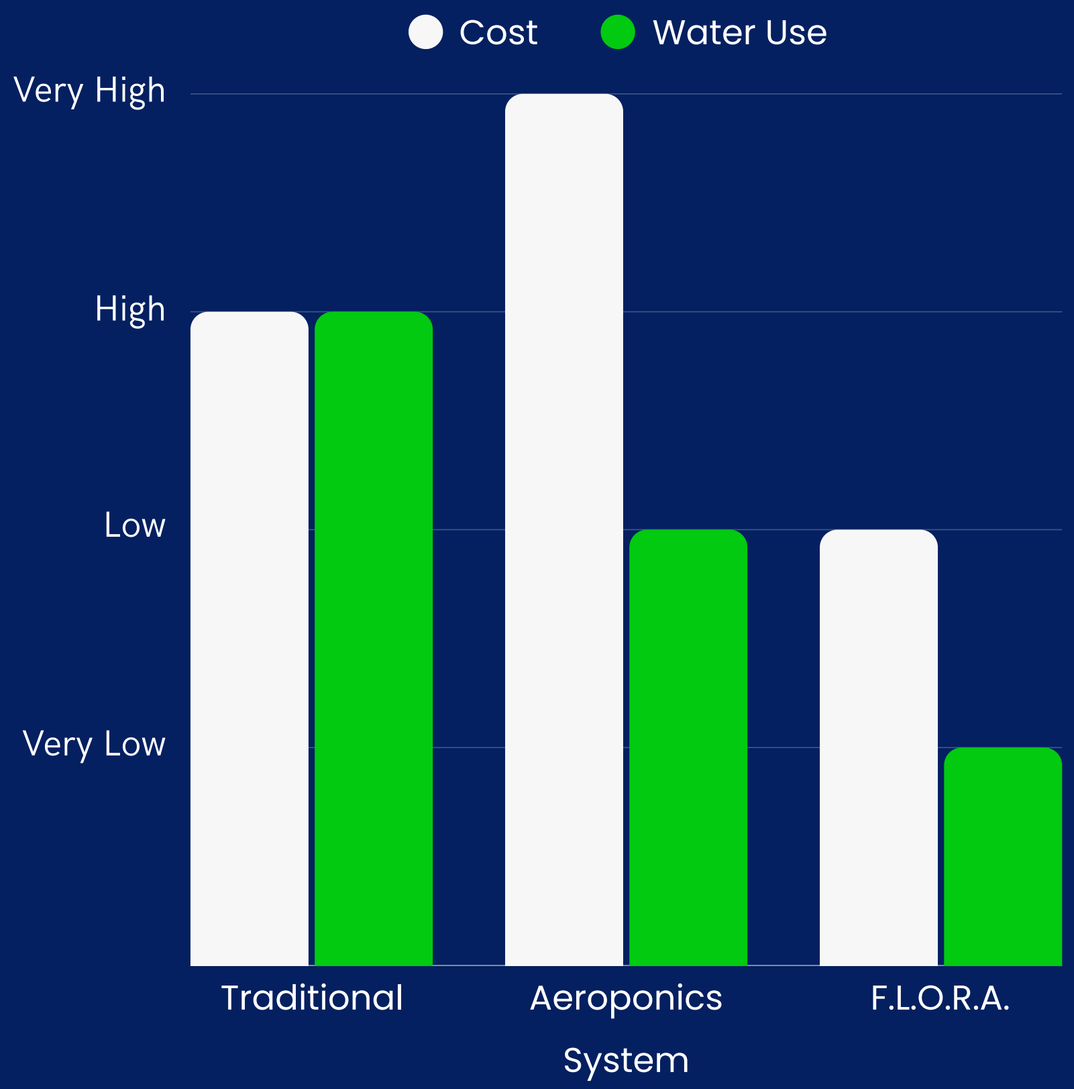
3. A live crop demonstration (featuring rapid-growth leafy greens like coriander) showcasing healthy, rot-free root development in a high-oxygen fog environment.





FEASIBILITY (SYSTEM DESIGN)

- Eliminates high-pressure nozzles, reducing clogging issues
- Uses a low-power mist maker (piezoelectric transducer)
- Mineral pebble layer filters mist without electricity
- No complex plumbing or moving parts → low maintenance
- Simple and reliable design suitable for real-world conditions



FEASIBILITY (COST & EFFICIENCY)

- Total power consumption is less than 20W, ideal for solar setups
- Low CapEx → affordable to build compared to traditional systems
- 80% reduction in fertilizer cost due to closed-loop system
- Minimal electricity usage → low operational cost (OpEx)
- Farmers can achieve fast ROI within a few crop cycles

SCALABILITY & USE CASES

- Modular design → start small and expand easily
- Can be scaled vertically using rack-based setups
- No need for expensive greenhouse infrastructure

- Urban farming in water-limited spaces
- Drought-prone regions with minimal water usage
- Cultivation of medicinal and high-value crops in a clean, soil-free environment

THANK YOU